

Vedant Desai

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EDUCATION

University of Michigan

B.S. in Computer Science and Economics

Expected May 2028

Ann Arbor, MI

- **GPA:** 3.66 / 4.0
- **Coursework:** Data Structures & Algorithms, Intro to Statistics and Data Analysis, Microeconomics, Macroeconomics, Discrete Mathematics, Calculus III, Physics (Mechanics)

EXPERIENCE

Vylet

May 2026 – Present | vyletdata.com

Founder — Automated lead-sourcing platform for PE/search-fund; live product generating \$1,500 MRR across three clients

- Launched Vylet, automating a ~30-minute manual process per business into a Dockerized LangGraph pipeline generating 30 scored leads in 30 minutes — a 30x speedup — with Redis/Celery workers on a recurring cycle.
- Engineered a LangSmith eval pipeline spanning 20 adversarial business test cases across 13 archetype labels — manufacturers, SaaS tools, PE holding companies, geographic mismatches — then layered deterministic Pydantic consensus gates to lift extraction faithfulness from 50% to 90%.
- Architected a custom asyncpg Data Access Layer storing Gemini embeddings alongside source records; engineered injection-safe SQL timestamp validation that detects stale entries and triggers automatic re-scrapes, keeping the lead database fresh without manual intervention.

Michigan Data Consulting (MDC)

Jan 2026 – May 2026

Data Engineer — Michigan Campaign Finance Network

Ann Arbor, MI

- Replaced manual Michigan campaign-finance research — portal searches capped by the Bureau of Elections, irregular Excel exports, hand normalization at ~2 hours per committee — with a Requests + Pandas ETL that ingests filings directly, eliminating ~800 hours of manual pulls across 400 tracked PACs.
- Delivered a production Flask REST API on AWS EC2 as the sole engineer on a 5-month MCFN contract, wiring ingested data and PAC rankings into the nonprofit's public-facing research workflow.

PROJECTS

SignalWeaver | *LangGraph, pgvector, MPNet, Docker, GitHub Actions*

github.com/Verdent06/SignalWeaver

Multi-signal financial research platform combining fundamentals, macro indicators, and news sentiment (research assistant; not investment advice)

- Implemented semantic search over stored financial news with pgvector cosine similarity (768-d MPNet embeddings), hitting 49ms p50 / 99ms p99 over 90 queries returning top-5 or top-10 matches.
- Orchestrated a 5-node LangGraph pipeline (fetch → classify → embed → score → explain) with per-node timing from the same 90-run batch: fetch 7.3s p50, classify 1.1s p50 (~29ms/article over 40-article batches), embed 1.1s p50, score under 1ms p50, explain 3ms p50 — fetch dominated wall time (~80% of p50 latency).
- Containerized the stack with Docker Compose (API + Postgres/pgvector + nginx frontend) and a GitHub Actions CI pipeline (frontend build, pytest, API image build on main).

Granular Synthesizer Plugin | *C++, JUCE*

github.com/Verdent06/granular-synth

Real-time audio DSP plugin built from scratch in C++/JUCE — sine oscillator through full granular engine with lock-free threading and release-ready VST3/AU binaries

- Enforced the audio thread's zero-allocation constraint by pre-allocating a MemoryPool<Grain, 64> slab per voice at plugin startup — a fixed free-list that acquires and releases grain slots without touching the heap — so processBlock() never calls new or delete after prepareToPlay() completes.

TECHNICAL SKILLS

Languages	Python, C++, SQL
AI / ML	LangGraph, LangSmith, Pandas
Frameworks	Flask, JUCE
Databases	Redis, pgvector
Cloud & Infrastructure	AWS (EC2), Docker, Celery, Git, GitHub Actions